

VESSEL ANALYSIS REPORT

vaf(%)	branch points count	tip point count	vein count	total vein length	average vein length
1.374	61.0	227.0	200.0	3912.762	19.564

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
1	[19,5]	[3,9]	21.243	BRANCH	BRANCH
2	[19,5]	[21,11]	5.828	BRANCH	BRANCH
3	[22,0]	[19,5]	4.828	TIP	BRANCH
4	[134,10]	[117,6]	17.657	TIP	BRANCH
5	[107,0]	[117,6]	11.071	TIP	BRANCH
6	[117,6]	[115,20]	13.414	BRANCH	BRANCH
7	[3,9]	[14,16]	12.899	BRANCH	BRANCH
8	[3,9]	[9,32]	28.728	BRANCH	BRANCH
9	[21,11]	[21,14]	2.0	BRANCH	BRANCH
10	[38,4]	[21,11]	19.314	TIP	BRANCH
11	[21,14]	[14,16]	6.828	BRANCH	BRANCH
12	[84,17]	[21,14]	81.113	TIP	BRANCH
13	[14,16]	[9,32]	19.071	BRANCH	BRANCH
14	[143,17]	[115,20]	88.912	TIP	BRANCH
15	[115,20]	[20,70]	139.296	BRANCH	BRANCH
16	[239,18]	[228,24]	12.485	TIP	BRANCH
17	[221,41]	[228,24]	19.485	TIP	BRANCH
18	[218,1]	[228,24]	28.556	TIP	BRANCH
19	[9,32]	[9,45]	12.0	BRANCH	BRANCH
20	[386,29]	[385,32]	2.414	TIP	BRANCH
21	[371,36]	[385,32]	14.657	TIP	BRANCH
22	[386,35]	[385,32]	2.0	TIP	BRANCH
23	[288,1]	[283,41]	41.071	TIP	BRANCH
24	[290,38]	[283,41]	7.828	TIP	BRANCH
25	[9,45]	[22,50]	14.071	BRANCH	BRANCH
26	[0,54]	[9,45]	12.899	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
27	[39,31]	[30,48]	20.314	TIP	BRANCH
28	[39,50]	[30,48]	8.828	TIP	BRANCH
29	[30,48]	[22,50]	7.828	BRANCH	BRANCH
30	[21,54]	[22,50]	3.414	TIP	BRANCH
31	[259,59]	[265,66]	9.071	TIP	BRANCH
32	[265,66]	[275,72]	11.485	BRANCH	BRANCH
33	[261,68]	[265,66]	3.414	TIP	BRANCH
34	[18,65]	[20,70]	4.414	TIP	BRANCH
35	[20,70]	[17,98]	37.627	BRANCH	BRANCH
36	[275,72]	[276,85]	12.414	BRANCH	BRANCH
37	[284,58]	[275,72]	16.314	TIP	BRANCH
38	[298,107]	[276,85]	32.456	TIP	BRANCH
39	[276,85]	[238,135]	65.497	BRANCH	BRANCH
40	[357,95]	[347,96]	9.414	TIP	BRANCH
41	[330,87]	[347,96]	19.728	TIP	BRANCH
42	[349,106]	[347,96]	9.828	TIP	BRANCH
43	[17,98]	[53,105]	37.899	BRANCH	BRANCH
44	[16,102]	[17,98]	3.0	TIP	BRANCH
45	[64,82]	[75,100]	22.728	TIP	BRANCH
46	[75,100]	[79,100]	3.0	BRANCH	BRANCH
47	[75,100]	[53,105]	22.657	BRANCH	BRANCH
48	[94,70]	[79,100]	35.213	TIP	BRANCH
49	[79,100]	[75,113]	13.657	BRANCH	BRANCH
50	[53,105]	[75,113]	25.485	BRANCH	BRANCH
51	[75,113]	[75,119]	5.0	BRANCH	BRANCH
52	[104,137]	[75,119]	36.627	TIP	BRANCH
53	[75,119]	[65,126]	11.485	BRANCH	BRANCH
54	[64,120]	[65,126]	5.414	TIP	BRANCH
55	[58,139]	[65,126]	14.899	TIP	BRANCH
56	[143,125]	[142,129]	3.414	TIP	BRANCH
57	[139,128]	[142,129]	2.414	TIP	BRANCH
58	[157,178]	[142,129]	55.87	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
59	[251,129]	[238,135]	14.485	TIP	BRANCH
60	[49,169]	[27,149]	53.385	TIP	BRANCH
61	[27,149]	[13,185]	40.799	BRANCH	BRANCH
62	[24,143]	[27,149]	5.828	TIP	BRANCH
63	[82,151]	[94,152]	13.657	TIP	BRANCH
64	[104,152]	[94,152]	9.0	TIP	BRANCH
65	[95,158]	[94,152]	5.414	TIP	BRANCH
66	[354,151]	[348,152]	5.414	TIP	BRANCH
67	[346,155]	[348,152]	2.828	TIP	BRANCH
68	[6,188]	[13,185]	7.243	TIP	BRANCH
69	[16,194]	[13,185]	9.243	TIP	BRANCH
70	[356,187]	[341,190]	15.243	TIP	BRANCH
71	[338,189]	[341,190]	2.414	TIP	BRANCH
72	[339,198]	[341,190]	7.828	TIP	BRANCH
73	[69,181]	[60,209]	31.556	TIP	BRANCH
74	[96,200]	[60,209]	40.385	TIP	BRANCH
75	[60,209]	[44,215]	17.071	BRANCH	BRANCH
76	[36,200]	[39,215]	15.243	TIP	BRANCH
77	[39,215]	[44,215]	4.0	BRANCH	BRANCH
78	[39,215]	[33,216]	5.414	BRANCH	BRANCH
79	[44,215]	[44,224]	8.0	BRANCH	BRANCH
80	[24,207]	[33,216]	11.899	TIP	BRANCH
81	[33,216]	[31,227]	10.414	BRANCH	BRANCH
82	[57,229]	[44,224]	14.071	TIP	BRANCH
83	[44,224]	[31,227]	13.243	BRANCH	BRANCH
84	[29,235]	[31,227]	7.414	TIP	BRANCH
85	[1,242]	[2,245]	2.414	TIP	BRANCH
86	[9,246]	[2,245]	6.414	TIP	BRANCH
87	[0,247]	[2,245]	1.414	TIP	BRANCH
88	[225,247]	[228,246]	2.414	TIP	BRANCH
89	[228,246]	[229,249]	2.414	BRANCH	BRANCH
90	[229,249]	[259,261]	38.071	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
91	[221,256]	[229,249]	9.485	TIP	BRANCH
92	[286,271]	[259,261]	31.899	TIP	BRANCH
93	[259,261]	[246,285]	28.971	BRANCH	BRANCH
94	[231,283]	[237,283]	5.0	BRANCH	BRANCH
95	[226,280]	[231,283]	4.828	TIP	BRANCH
96	[229,286]	[231,283]	2.414	TIP	BRANCH
97	[237,275]	[237,283]	7.0	TIP	BRANCH
98	[237,283]	[246,285]	8.414	BRANCH	BRANCH
99	[246,285]	[265,303]	28.385	BRANCH	BRANCH
100	[219,295]	[217,296]	1.414	TIP	BRANCH
101	[199,311]	[217,296]	24.971	TIP	BRANCH
102	[215,294]	[217,296]	1.414	TIP	BRANCH
103	[272,300]	[265,303]	7.243	TIP	BRANCH
104	[269,312]	[265,303]	9.657	TIP	BRANCH
105	[88,309]	[88,320]	10.0	TIP	BRANCH
106	[123,286]	[88,320]	56.284	TIP	BRANCH
107	[88,320]	[82,331]	12.485	BRANCH	BRANCH
108	[326,314]	[336,320]	13.243	TIP	BRANCH
109	[329,326]	[336,320]	8.071	TIP	BRANCH
110	[62,324]	[82,331]	23.071	TIP	BRANCH
111	[78,354]	[82,331]	23.657	TIP	BRANCH
112	[39,381]	[12,337]	63.941	TIP	BRANCH
113	[30,353]	[12,337]	26.556	TIP	BRANCH
114	[34,317]	[12,337]	57.213	TIP	BRANCH
115	[66,372]	[64,388]	16.657	TIP	BRANCH
116	[51,385]	[64,388]	13.243	TIP	BRANCH
117	[85,413]	[64,388]	35.213	TIP	BRANCH
118	[67,434]	[67,441]	6.0	TIP	BRANCH
119	[75,443]	[67,441]	7.828	TIP	BRANCH
120	[41,451]	[67,441]	29.142	TIP	BRANCH
121	[155,432]	[149,444]	13.485	TIP	BRANCH
122	[152,445]	[149,444]	2.414	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
123	[146,445]	[149,444]	2.414	TIP	BRANCH
124	[180,442]	[180,444]	1.0	TIP	BRANCH
125	[175,447]	[180,444]	5.243	TIP	BRANCH
126	[184,455]	[180,444]	11.243	TIP	BRANCH
127	[35,454]	[34,460]	5.414	TIP	BRANCH
128	[39,460]	[34,460]	4.0	TIP	BRANCH
129	[18,483]	[34,460]	28.213	TIP	BRANCH
130	[55,461]	[70,475]	20.971	TIP	BRANCH
131	[70,475]	[83,479]	13.657	BRANCH	BRANCH
132	[63,486]	[70,475]	12.485	TIP	BRANCH
133	[91,473]	[83,479]	9.485	TIP	BRANCH
134	[101,509]	[83,479]	37.042	TIP	BRANCH
135	[371,470]	[377,488]	19.485	TIP	BRANCH
136	[381,488]	[377,488]	3.0	TIP	BRANCH
137	[375,506]	[377,488]	17.828	TIP	BRANCH
138	[157,490]	[151,505]	16.485	TIP	BRANCH
139	[161,510]	[151,505]	12.243	TIP	BRANCH
140	[371,1]	[367,11]	11.657	TIP	TIP
141	[186,7]	[198,22]	23.728	TIP	TIP
142	[478,11]	[456,29]	31.213	TIP	TIP
143	[310,12]	[307,28]	19.485	TIP	TIP
144	[411,13]	[417,32]	23.142	TIP	TIP
145	[59,33]	[64,46]	22.728	TIP	TIP
146	[337,39]	[351,48]	17.728	TIP	TIP
147	[490,40]	[474,41]	17.243	TIP	TIP
148	[191,44]	[212,46]	22.657	TIP	TIP
149	[168,48]	[178,52]	13.071	TIP	TIP
150	[235,56]	[183,82]	72.083	TIP	TIP
151	[124,57]	[122,71]	16.485	TIP	TIP
152	[255,65]	[241,66]	14.414	TIP	TIP
153	[147,75]	[136,87]	17.142	TIP	TIP
154	[466,78]	[463,97]	21.071	TIP	TIP

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
155	[214,86]	[185,92]	31.485	TIP	TIP
156	[307,87]	[309,96]	10.657	TIP	TIP
157	[440,88]	[446,111]	28.556	TIP	TIP
158	[165,94]	[148,102]	20.899	TIP	TIP
159	[116,103]	[102,121]	32.485	TIP	TIP
160	[201,109]	[206,110]	6.243	TIP	TIP
161	[317,114]	[289,128]	34.971	TIP	TIP
162	[129,124]	[117,137]	21.142	TIP	TIP
163	[376,126]	[381,142]	22.899	TIP	TIP
164	[5,140]	[0,149]	11.071	TIP	TIP
165	[204,148]	[209,157]	11.071	TIP	TIP
166	[290,160]	[325,188]	51.284	TIP	TIP
167	[179,174]	[197,188]	23.799	TIP	TIP
168	[431,178]	[437,192]	22.485	TIP	TIP
169	[245,187]	[257,192]	15.243	TIP	TIP
170	[135,193]	[138,210]	19.071	TIP	TIP
171	[177,196]	[170,198]	7.828	TIP	TIP
172	[136,224]	[102,226]	38.971	TIP	TIP
173	[171,234]	[179,236]	8.828	TIP	TIP
174	[185,236]	[199,241]	16.071	TIP	TIP
175	[151,238]	[159,248]	13.899	TIP	TIP
176	[217,254]	[213,271]	19.243	TIP	TIP
177	[70,259]	[95,261]	29.971	TIP	TIP
178	[181,269]	[200,273]	21.485	TIP	TIP
179	[291,278]	[286,286]	11.828	TIP	TIP
180	[7,289]	[55,312]	72.355	TIP	TIP
181	[255,311]	[246,322]	14.728	TIP	TIP
182	[365,318]	[377,332]	25.314	TIP	TIP
183	[237,337]	[237,349]	12.828	TIP	TIP
184	[136,348]	[150,355]	16.899	TIP	TIP
185	[169,356]	[154,357]	16.243	TIP	TIP
186	[243,360]	[242,383]	31.799	TIP	TIP

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
187	[402,362]	[358,367]	48.556	TIP	TIP
188	[314,363]	[307,364]	8.243	TIP	TIP
189	[446,365]	[425,417]	60.698	TIP	TIP
190	[296,371]	[280,381]	23.556	TIP	TIP
191	[129,374]	[130,388]	16.657	TIP	TIP
192	[166,397]	[153,409]	17.971	TIP	TIP
193	[119,411]	[86,420]	37.556	TIP	TIP
194	[31,415]	[49,430]	24.799	TIP	TIP
195	[225,423]	[191,424]	36.071	TIP	TIP
196	[12,435]	[12,442]	7.828	TIP	TIP
197	[118,443]	[142,450]	27.728	TIP	TIP
198	[452,469]	[431,474]	23.899	TIP	TIP
199	[278,494]	[308,511]	39.385	TIP	TIP
200	[60,500]	[39,510]	29.385	TIP	TIP